

ECHINODERMS

Echinoderms include an astonishing array of sea creatures, ranging from filter-feeding feather stars to grazing urchins and predatory starfish.

This is the only large invertebrate phylum entirely restricted to salt water. Echinoderms are sluggish, slow-moving inhabitants of the sea floor and most have a body form built around a five-part radial symmetry. Echinoderm means “spiny-skinned” – a reference to the hard, calcified structures, called ossicles, that make up the internal skeleton of these animals. In starfish these are spaced out within the soft tissue so that the animal is reasonably flexible, but in urchins they are fused together to form a solid internal shell. In sea cucumbers the ossicles are so minute and sparsely distributed – or even absent – that the entire animal is soft-bodied.

HYDRAULIC FEET

Echinoderms are the only animals with a water transport system. Sea water is drawn into the centre of the body through a perforated, sieve-like plate, usually on the upper surface of the body.

The water circulates through tubes that run through the animal and is forced into tiny soft projections near the surface, called tube feet, which can move back and forth and stick to surfaces. In feather stars these tube feet point upwards on the feathery arms, capturing particles of food, which are then moved towards the central mouth. In other echinoderms thousands of downward-pointing tube feet are used in locomotion – together they pull the animal over sediment and rock.

DEFENCE

The tough prickly skin of most echinoderms offers good protection from potential predators, but these animals defend themselves in other ways too. Many urchins are also covered with formidable spines; some can inflict serious wounds in humans. Many species have tiny pincer-like projections, sometimes venomous, that can remove cluttering debris as well as deter potential predators. Soft-bodied sea cucumbers rely on noxious chemicals to protect them – and are often brightly coloured as a warning. As a last resort, some sea cucumbers can even spew out sticky tangles or eject their gut.

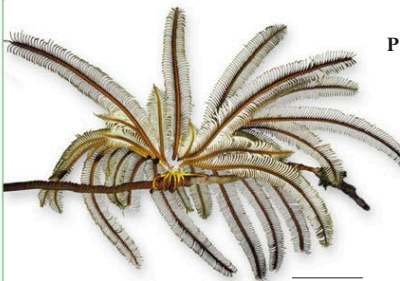
PHYLUM	ECHINODERMATA
CLASSES	5
ORDERS	35
FAMILIES	173
SPECIES	7,447



The tiny tube feet on which echinoderms move around the seabed are seen here on a crown-of-thorns starfish, *Acanthaster planci*.

FEATHER STARS

A feather star carries five basic filter-feeding arms, which in some species branch into a dense cluster. A mouth and anus point upwards from the centre of the star. Some members of the class Crinoidea crawl along on their feather-like arms. Others, known as sea lilies, are attached by a stalk.

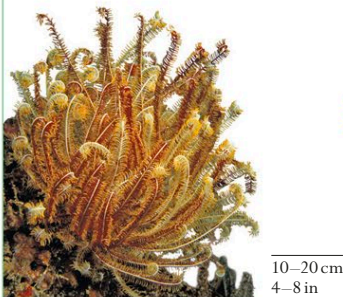


10–15 cm
4–6 in

PRETTY FEATHER STAR

Cenometra emendatrix
F: Colobometridae

The arms of a feather star are fringed with smaller branches called pinnules that catch particles of food more effectively. In this tropical Pacific species they are white.

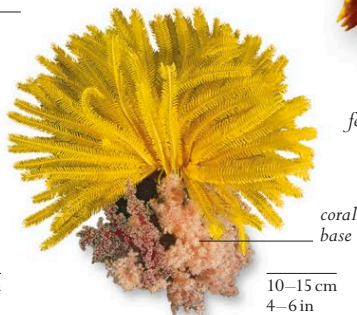


10–20 cm
4–8 in

GOLDEN FEATHER STAR

Davidaster rubiginosus
F: Comatulidae

Like other crinoids, this species from the W. Atlantic, is a suspension feeder, catching plankton with its feathery arms.

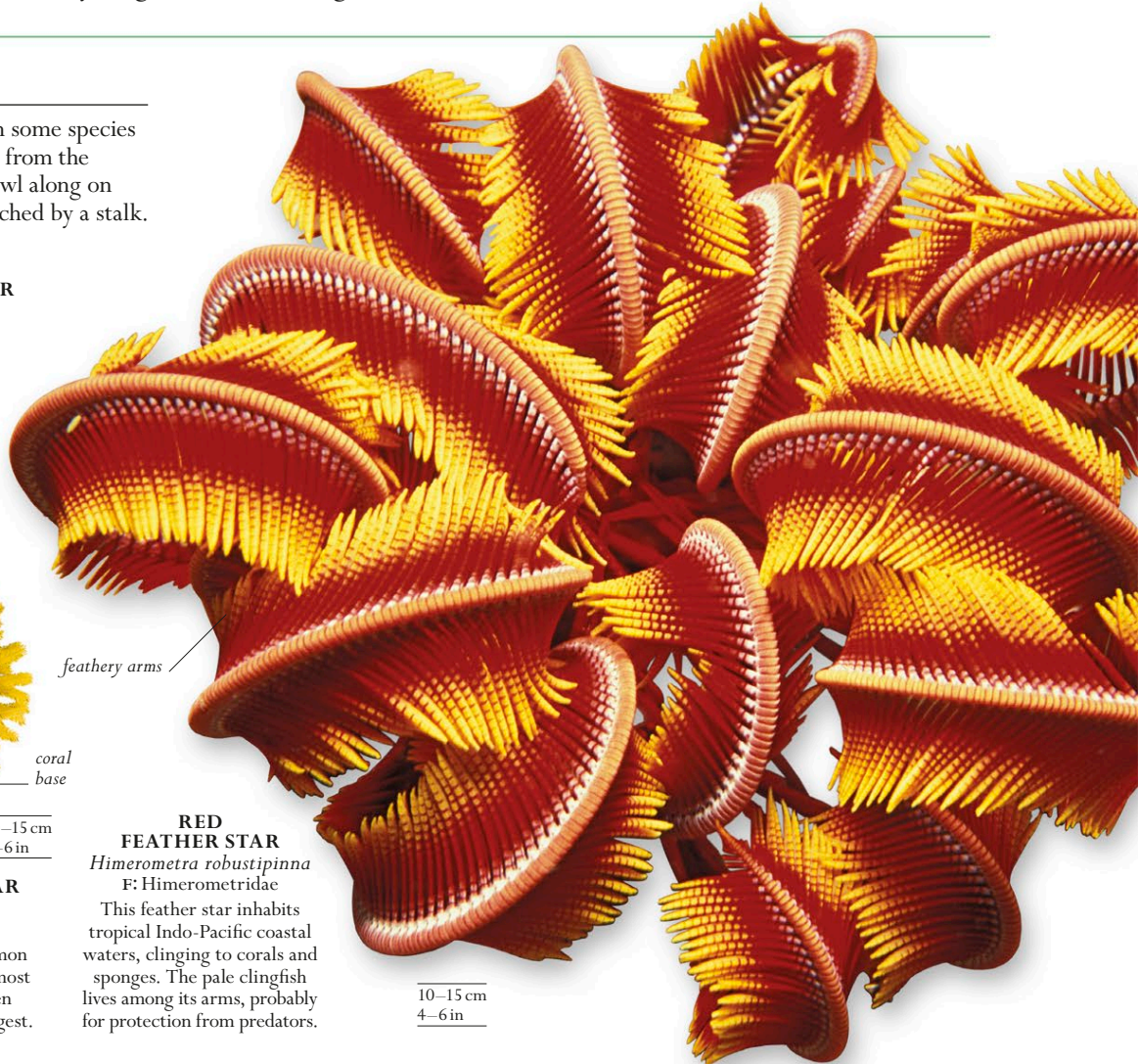


10–15 cm
4–6 in

YELLOW FEATHER STAR

Anneissia bennetti
F: Comasteridae

Like many crinoids, this common W. Pacific species filter feeds most voraciously at night and when underwater currents are strongest.



RED FEATHER STAR

Himerometra robustipinna
F: Himerometridae

This feather star inhabits tropical Indo-Pacific coastal waters, clinging to corals and sponges. The pale clingfish lives among its arms, probably for protection from predators.

10–15 cm
4–6 in